

Simultaneous avoidance of countable families

21 September 2026 — continuation for discussion

Local theorem, with a hand proof below. A countable family of sequences over a fixed finite alphabet has a single avoiding population if and only if none of its members is eventually periodic. For an alphabet of size $n \geq 2$, the population can have n roots and at most n children per vertex, which is the smallest possible uniform outdegree bound. If the sequences have a uniformly computable enumeration and are all non-eventually-periodic, such a population can be constructed computably. Independent review and external priority checking remain; this addition has not been checked in Lean.

The earlier explicit P_s construction avoided one prescribed sequence; `complement-and-barriers.md` proves that it also avoids its complement. The present result treats arbitrary countable families, using a different label sequence built by successive finite extensions.

1. A common space of populations

For any $r \in \{0,1\}^{\mathbb{N}}$, let G_r have vertices $\mathbb{N}$, birthdate v at v , roots 0 and 1, and precisely these incoming edges for each $v \geq 2$:

$$v-1 \longrightarrow v \text{ labelled } r(v), \quad v-2 \longrightarrow v \text{ labelled } 1-r(v).$$

There is no edge $0 \rightarrow 1$. Every G_r satisfies the population axioms: infinite vertex set, finitely many roots, finite birthdate sublevels, birthdates increasing along edges, one incoming parent of each label at every non-root, and outdegree at most two. Every infinite path has increments one or two. The values $r(0), r(1)$ do not label edges but are harmless coordinates.

2. Finite-extension lemma

Lemma. Fix a non-eventually-periodic binary sequence s and any prescribed prefix $r(0), \dots, r(N)$, with $N \geq 1$. There is an extension r for which G_r avoids s from every starting vertex.

Set $C = -N - 2$ and prescribe the unassigned tail by

$$r(v) = s\left(\left\lfloor \frac{v-C}{2} \right\rfloor\right) \text{ xor } ((v-C) \bmod 2), \quad v > N.$$

All indices on the right are nonnegative. The old prefix is unchanged. Suppose there were an infinite s -matching path $v_0, v_1, \dots$ and put $d_k = v_k - 2k$. Each step changes d_k by -1 or 0 .

Let j be its first index with $v_j > N$; this exists because vertices strictly increase. If $j = 0$, then $d_j = v_0 \geq 0 > C$. Otherwise $v_j \leq N + 2$ because edges have length at most two. Also $j \leq v_j$ because $v_0 \geq 0$ and every increment is at least one. Consequently

$$d_j = v_j - 2j \geq -v_j \geq -N - 2 = C.$$

For $k \geq j$, the lower bound $d_k \geq C$ is invariant. At equality, $v_k = 2k + C$ and an increment-one edge would have label

$$r(v_k + 1) = s(k) \text{ xor } 1,$$

so it cannot match. If $d_k > C$, decreasing by at most one preserves the bound. The edge endpoints used here are beyond N , so the prescribed tail formula really applies; the arbitrary old prefix is no longer involved.

The bounded nonincreasing integer d_k therefore stabilizes at some $D \geq C$. All subsequent increments are two. Write $h = D - C \geq 0$ and $q = \lfloor h/2 \rfloor + 1 > 0$. Matching the increment-two edge says

$$s(k) = 1 - r(2k + D + 2).$$

For even h this is $s(k) = 1 - s(k + q)$; for odd h it is $s(k) = s(k + q)$. These give eventual periods $2q$ and q , respectively, contradicting the hypothesis on s . This proves the lemma, including arbitrary starts and arbitrary preserved finite prefixes.

3. Simultaneous avoidance

For a fixed target s and start i , define

$$U_{s,i} = \{r : G_r \text{ has no infinite } s\text{-matching path from } i\}.$$

This set is open in Cantor space. The matching-prefix tree is finitely branching. By König's lemma, absence of an infinite branch means that some finite matching level is empty. A level of depth b only reads labels at vertices at most $i + 2b$, so its emptiness is preserved by fixing a finite prefix of r .

If s is non-eventually-periodic, $U_{s,i}$ is dense: any finite cylinder can first be extended through coordinate 1, then extended by the preceding lemma to a member of $U_{s,i}$.

Now let $(s_j)_{j \in J}$ be a countable family of non-eventually-periodic binary sequences. The set of simultaneous avoiders is

$$\bigcap_{j \in J} \bigcap_{i \in \mathbb{N}} U_{s_j, i}.$$

It is a countable intersection of dense open sets in Cantor space, hence comeagre and nonempty by the Baire category theorem. Any member gives the required single two-root, degree-two population. An empty family is trivial.

Conversely, a family containing an eventually periodic member cannot have an avoiding population, by Alexander's eventual-periodic unavoidability theorem (with the boundary repair recorded in `results.md`). This proves the stated classification for countable binary families.

This is a category statement about the choice of **edge labels** r , not an almost-sure statement under random labels. No measure-one claim is made.

4. Effective construction under a uniform promise

The Baire proof can be implemented directly. Assume $s_j(k)$ is uniformly computable in (j, k) , with the promise that every s_j is non-eventually periodic. Enumerate all requirements (j, i) , and maintain a finite prefix of r extending through at least coordinate 1.

At a stage for (j, i) :

1. Form the lemma's temporary infinite extension of the current prefix using target s_j .
2. Enumerate the finite matching levels from i in this temporary population until the first empty level. The lemma and König's lemma guarantee that this search terminates.
3. Freeze all bits of that extension read by the search, together with at least one further coordinate beyond the old prefix.

The finite barrier for this requirement persists under every later extension. Every requirement is visited, and the prefix grows at every stage. To compute one desired output bit, run stages until that bit has been frozen. Each stage terminates and uses finitely many values of the uniformly computable targets, so the final r is computable. The same construction works relative to an oracle presenting an arbitrary sequence of targets.

No algorithm recognizing the promise is asserted. There is no contradiction with taking the countable family of **all computable aperiodic sequences**: the existence theorem handles that family, but it has no uniformly computable enumeration containing only aperiodic sequences and covering them all. Indeed, such an enumeration would yield a computable r avoiding all of them. If r were aperiodic, the increment-one path $1, 2, 3, \dots$ would realize the computable aperiodic sequence $r(2), r(3), \dots$, a contradiction. If r were eventually periodic, that same path realizes an eventually periodic word but does not by itself settle the contradiction; use instead the next paragraph's periodic-tail argument.

If r is eventually periodic with period p , G_r realizes every binary sequence. For any desired finite word, choose its endpoint sufficiently far past the exceptional prefix and follow its uniquely labelled incoming parents backwards. All vertices of this finite path can be kept in the periodic tail. Translate the path backwards by a multiple of p until its start lies in a fixed tail interval of p vertices. The edge labels remain the same and all vertices remain in the tail. Arbitrarily long prescribed prefixes are thus realized from this finite set of starts. König's lemma gives an infinite realizing path. This proves universality and completes the contradiction for eventually periodic r as well.

5. Optimal degree for every finite alphabet

Identify an alphabet of size $n \geq 2$ with $\mathbb{Z}/n\mathbb{Z}$. Use vertices $\mathbb{N}$, roots $0, \dots, n-1$, birthdate v , and, for each $v \geq n$, incoming edges $v - a \rightarrow v$, $1 \leq a \leq n$. At each endpoint assign the n labels bijectively to these edges. All such assignments give populations of outdegree at most n . The parameter space is a product of finite permutation spaces, one for each endpoint, and is a Baire space.

General finite-extension lemma. Preserve arbitrary incoming-label permutations at endpoints through $N \geq n - 1$. For a fixed aperiodic target s , set $C = -(n - 1)(N + n)$ and label all later edges by

$$\ell(v - a, v) = s\left(\left\lfloor \frac{v - C}{n} \right\rfloor\right) + a - ((v - C) \bmod n) - 1 \pmod{n}, \quad v > N.$$

As a ranges from 1 to n these labels are a permutation, as required. Suppose v_k were an infinite matching path and put $d_k = v_k - nk$. Its increments lie in $\{1 - n, \dots, 0\}$. At the first entry j into vertices beyond N , either $j = 0$ and $d_j \geq 0$, or $v_j \leq N + n$ and $j \leq v_j$, giving

$$d_j \geq -(n - 1)v_j \geq -(n - 1)(N + n) = C.$$

To check that $d_k \geq C$ persists, set $h = d_k - C \geq 0$. A step of length a could cross below C only if $h + a < n$. In that case $0 \leq h \leq n - 2$, the endpoint is $v_{k+1} = nk + C + h + a$, and its edge label is

$$s(k) + a - (h + a) - 1 = s(k) - h - 1 \pmod{n}.$$

Since $1 \leq h + 1 \leq n - 1$, this differs from $s(k)$. Such a crossing cannot be matching. All relevant endpoints are beyond N , so the tail formula applies.

Thus d_k eventually stabilizes at $D \geq C$ and every later edge has length n . Write $h = D - C$, $b = h \bmod n$ and $q = \lfloor h/n \rfloor + 1 > 0$. Matching then requires

$$s(k) = s(k + q) + n - b - 1 \pmod{n}$$

for all sufficiently large k . Iterating n times gives the positive eventual period nq , a contradiction. This proves the lemma.

Absence of a matching path from a fixed start is again open (depth b reads only endpoints through $i + nb$) and dense by this lemma. The same countable intersection and staged freezing arguments prove simultaneous avoidance and its effective version with n roots and maximum outdegree n .

For a single target there is an even simpler formula: use the displayed label rule with $C = 0$ at *every* endpoint $v \geq n$. The invariant starts at $v_0 \geq 0$, and the same proof applies immediately. This generalizes the original binary construction directly, without a binary code or vertex-copy degree overhead.

Optimality of the degree bound. In any n -label population with R roots and maximum outdegree at most $n - 1$, take a finite birthdate sublevel set F containing all roots and M vertices. It is ancestor-closed. The incoming-label requirement gives at least $n(M - R)$ internal edges, while the outdegree assumption gives at most $(n - 1)M$. Hence $M \leq nR$. Birthdate sublevels have

unbounded size in an infinite population, a contradiction. Thus no such population exists below degree n at all. This counting obstruction was already recorded in the earlier supplement; the new point is attainment for arbitrary countable aperiodic target families. The one-symbol case has no aperiodic targets; an empty family admits a chain.

6. What is now settled, and what is not

- Arbitrary finite collections of aperiodic targets over a fixed finite alphabet can be avoided simultaneously; the result extends to countable collections. The smallest possible uniform outdegree is attained for every finite alphabet.
- In particular one population can avoid every shift and complement of Thue–Morse. It must nevertheless realize some other member of the Thue–Morse orbit closure, by the earlier subshift-intersection theorem.
- There is a population avoiding every computable aperiodic binary sequence. For the present G_r family, its labelling cannot be computable, as just shown. No claim about every other computable population presentation is made.
- All uncountable families cannot be handled in this way: the family of all aperiodic binary sequences is unavoidable as a set, since every binary population realizes continuum many aperiodic words. Countability is a sufficient hypothesis, not a proved necessary one for joint avoidance.

The remaining questions include a characterization of jointly avoidable uncountable families, quantitative costs of the effective finite-extension construction, sharp Thue–Morse matching barriers, and the other embedding/ordinal variants in the original packet. None is silently resolved by the countable intersection.

Finite controls in `evidence/44u-countable-avoidance-checks.py` check arbitrary short preserved prefixes, the shifted-offset boundary, and a small finite initial segment of the staged construction. They cannot certify countably many requirements; that conclusion uses the proofs above.

Source: Samuel A. Alexander, *Biologically unavoidable sequences*, Electronic Journal of Combinatorics **20**(1) (2013), P31, doi:10.37236/3035. The finite-extension lemma and its use here are local continuations of the earlier proposed classification proof. Baire category and König’s lemma are standard antecedents, not new results claimed by this note.